

-- Cox Regression - Sarcopenia Onset -----

i Definition : ewgsop2
i Covariates : time_dependent
i Dairy : continuous
i Route : mice
i Interaction: none

-- MICE route - pooling across imputed datasets --

Detected 20 imputed datasets.
Fitting .imp = 1 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 49 events | 7801.6 person-years
Fitting .imp = 2 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 51 events | 7801.6 person-years
Fitting .imp = 3 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1949 intervals | 52 events | 7804.1 person-years
Fitting .imp = 6 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 51 events | 7801.6 person-years
Fitting .imp = 7 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years
Fitting .imp = 8 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 49 events | 7801.6 person-years
Fitting .imp = 9 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years
Fitting .imp = 10 / 20 ...
Events: 58 sarcopenia cases out of 1121 participants

??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years
Fitting .imp = 12 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years
Fitting .imp = 13 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 51 events | 7801.6 person-years
Fitting .imp = 14 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years
Fitting .imp = 15 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years
Fitting .imp = 16 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

```
Fitting .imp = 18 / 20 ...  
Events: 55 sarcopenia cases out of 1121 participants
```

```
-- Time-dependent covariates dataset (one row per pt x visit) --
```

```
??? Time-dependent dataset: 1948 intervals | 49 events | 7801.6 person-years  
Fitting .imp = 19 / 20 ...  
Events: 55 sarcopenia cases out of 1121 participants
```

```
-- Time-dependent covariates dataset (one row per pt x visit) --
```

```
??? Time-dependent dataset: 1948 intervals | 49 events | 7801.6 person-years  
Fitting .imp = 20 / 20 ...  
Events: 56 sarcopenia cases out of 1121 participants
```

```
-- Time-dependent covariates dataset (one row per pt x visit) --
```

```
??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years
```

```
-- Pooled adjusted model results (Rubin's rules) --
```

```
# A tibble: 8 x 5 term HR CI_low CI_high p.value <chr> <dbl> <dbl> <dbl> <dbl>  
1 dairy_exposure 1.00 0.999 1.00 0.649 2 BMI 0.712 0.636 0.796 0.000000391 3  
education_levelMedium (ISCED 3-4) 0.893 0.390 2.05 0.784 4 education_levelHigh  
(ISCED 5-8) 0.915 0.296 2.83 0.874 5 smoking_statusFormer 0.895 0.479 1.67  
0.721 6 smoking_statusCurrent 0.766 0.278 2.11 0.598 7 mvpa_min_day_f1 0.999  
0.996 1.00 0.459 8 diabetes_statusDiabetes 1.69 0.370 7.68 0.490
```

-- 1. Proportional Hazards - cox.zph() | pass: global p > 0.05

```
chisq df p dairy_exposure 1.1061 1 0.29 BMI 0.0878 1 0.77 education_level
3.2194 2 0.20 smoking_status 3.8324 2 0.15 mvpa_min_day_f1 0.2305 1 0.63
diabetes_status 2.1380 1 0.14 GLOBAL 10.4661 8 0.23
??? PH assumption not violated (global p = 0.2338)
Schoenfeld plots saved to
03_outputs/Cox/mice_ewgsop2_time_dependent_continuous_27072026_1259
```

-- 2. Linearity - martingale (null model) vs continuous predictors | pass: smooth ??? linear

```
'geom_smooth()' using formula = 'y ~ x'
'geom_smooth()' using formula = 'y ~ x'
'geom_smooth()' using formula = 'y ~ x'
Linearity plots saved to
03_outputs/Cox/mice_ewgsop2_time_dependent_continuous_27072026_1259
```

-- 3. Outliers - deviance & martingale residuals | pass: |dev| <= 3

i 1 observation(s) with |deviance residual| > 3.

-- 4. Influential observations - std. DFBetas | pass: no extreme spikes

i 26 obs exceed |DFBeta_std| > 0.2.

-- 5. Overall Fit - Cox-Snell residuals | pass: points on 45° diagonal

```
Cox-Snell plot saved to
03_outputs/Cox/mice_ewgsop2_time_dependent_continuous_27072026_1259
```

-- 6. Multicollinearity - VIF | pass: VIF < 5 (< 10 relaxed)

```
GVIF Df GVIF^(1/(2*Df)) dairy_exposure 1.063 1 1.031 BMI 1.132 1 1.064
education_level 1.121 2 1.029 smoking_status 1.049 2 1.012 mvpa_min_day_f1
1.043 1 1.021 diabetes_status 1.073 1 1.036
```

-- Adjusted scenario survival curves (time-dependent) [mice_imp1]

-- Incidence rates by dairy quartile [mice]

quartile	n_pts	n_events	person_years	rate_per_1000py	ci_low	ci_high
1 Q1 (low)	281	10.00	1965.30	5.09	2.44	9.36
2 Q2	280	9.70	1967.92	4.93	2.33	9.15
3 Q3	280	13.00	1934.70	6.72	3.58	11.49
4 Q4 (high)	280	17.45	1933.80	9.02	5.30	14.36

Incidence table saved to

03_outputs/Cox/mice_ewgsop2_time_dependent_continuous_27072026_1259

Model results saved to

03_outputs/Cox/mice_ewgsop2_time_dependent_continuous_27072026_1259/mice_model_results.csv